LEARNING SESSION S8

Tools for Data Science & AI

ChatGPT, Claude, Gemini, Copilot, Grok, Perplexity, Lovable, NotebookLM, Cursor and the core data stack: what each is for and how to use it

AnalytixLabs · Advanced Certification in Data Science & AI (IIT), June 2026 batch

Interactive version (every tab clickable):
https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw

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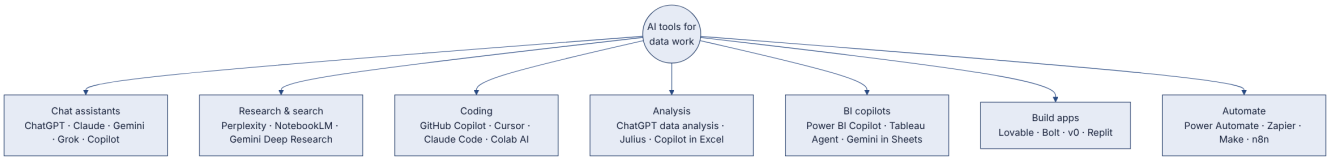
🧰 Tools for Data Science & AI

ChatGPT, Claude, Gemini, Copilot, Grok, Perplexity, Lovable, NotebookLM, Cursor and the core data stack: what each is for and how to use it

Learning session · tool landscape as of October 2026 · features and plans change often, so check each vendor's site

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Tool map



Seven groups of AI tools an analyst uses.

Which tool when

I need to…	First choice	Alternatives
Write/fix an Excel formula, SQL or Python	ChatGPT or Claude	Copilot in Excel, Gemini, GitHub Copilot
Analyse an uploaded CSV/Excel and make charts	ChatGPT data analysis	Claude (analysis + artifacts), Julius AI, Copilot in Excel, Gemini in Sheets
Research a topic with sources	Perplexity	ChatGPT / Gemini deep research, Claude research, Grok DeepSearch
Study from my own PDFs and notes	NotebookLM	Claude Projects, ChatGPT Projects
Real-time public sentiment / news on X	Grok	Perplexity
Build a small web app or dashboard without coding	Lovable	Bolt, v0, Replit, Claude artifacts
Write production code in an editor	GitHub Copilot or Cursor	Claude Code, Gemini in Colab
AI inside Office documents	Microsoft 365 Copilot	Gemini in Google Workspace
Automate workflows between apps	Power Automate (Microsoft shops)	Zapier, Make, n8n
Slides from an outline	Copilot in PowerPoint / Gamma	Canva Magic, Claude/ChatGPT drafts

AI assistants

Tool (maker)	Best at	Data-science uses	Notes
ChatGPT (OpenAI)	All-round assistant; runs Python on uploaded files; image generation; voice; custom GPTs; agent mode	Upload sales.xlsx → "clean it, show monthly trend, test if Pune differs from Delhi"; write SQL/DAX; explain errors	Free tier; Plus/Pro/Team/Enterprise paid plans. The course's GenAI classes used ChatGPT 3.5/4.
Claude (Anthropic)	Long documents, careful reasoning and writing, coding; Artifacts (interactive pages/dashboards); Projects with your files; Claude Code for agentic coding	Read a 100-page report and extract KPIs; build an interactive dashboard as an artifact; refactor a Python pipeline	Free, Pro, Max, Team, Enterprise plans. Also available inside Excel's Agent Mode and via API.
Gemini (Google)	Google Workspace integration (Sheets, Docs, Gmail, Drive), multimodal input, Deep Research	Gemini in Sheets: formulas, tables, analysis; Gemini in Colab: write and fix notebook code	Free tier; Google AI plans; Workspace business plans
Microsoft Copilot	Microsoft 365 apps: Excel (formulas, analysis, Agent Mode, `=COPILOT()`), PowerPoint, Word, Teams meeting recaps; Analyst and Researcher agents	"Analyse this table and add a pivot of sales by region"; turn Excel findings into a PowerPoint	Free Copilot app; Microsoft 365 Copilot licence for work data
Grok (xAI)	Real-time information from X (Twitter) and the web; DeepSearch; witty conversational style	Track brand sentiment or trending topics; quick current-events context	Available in X and the Grok app; free and paid tiers. Verify claims; social posts are noisy.
Perplexity	Answer engine with citations for every claim; Pro/Deep Research; Spaces for collections; Comet AI browser	Market sizing, competitor research, finding datasets and documentation with sources	Free and Pro; best for research, not for heavy file analysis
Others	Meta AI (in WhatsApp/Instagram), DeepSeek and Qwen (open-weight models), Mistral (European), Microsoft Copilot Studio (build your own agents)	Low-cost or self-hosted alternatives	Check data-privacy terms before uploading work data

One task, different assistants: prompt templates

```
ANALYSE (ChatGPT / Claude / Julius)
"You are a data analyst. The attached file has one row per order (columns: order_id, date,
city, category, amount). 1) Check data quality and list issues. 2) Monthly revenue trend
chart. 3) Top 5 cities and their share. 4) Three insights for a sales manager in plain English.
Show the code you ran."
```

```
RESEARCH (Perplexity)
"Size of India's quick-commerce market in 2025–26 with sources; list 5 credible reports and
their estimates in a table with publication dates."

LEARN (NotebookLM)
Upload: course PDFs and your notes → "Make a 20-question quiz on hypothesis testing and an
audio overview I can listen to on my commute."

BUILD (Lovable)
"Build a web app where a store manager uploads a CSV of daily sales and sees KPI cards, a
monthly trend chart and a top-products table. Clean, mobile-friendly design."
```

Build, code & research

Tool	What it is	Use it for
Lovable	AI app builder: describe an app in chat and it generates a full-stack web app (React front end, Supabase database/auth), with GitHub sync and one-click hosting	Prototypes, internal tools, a portfolio app (e.g. a churn-risk calculator) without writing code
Bolt, v0 (Vercel), Replit	Similar prompt-to-app builders; v0 focuses on UI; Replit has a full cloud IDE and agent	Quick front ends, dashboards, demos
NotebookLM (Google)	Research notebook grounded only in sources you upload; cites passages; generates study guides, mind maps and audio/video overviews	Revising course material, summarising long reports
GitHub Copilot	AI pair programmer inside VS Code, JetBrains, GitHub; chat and agent modes	Python, SQL, notebooks, unit tests
Cursor	AI-first code editor (VS Code-based) that edits across a whole project	Larger refactors, building pipelines
Claude Code	Agentic coding tool that reads a codebase, runs commands and makes changes	Automating data pipelines, building apps end to end
Julius AI	Chat-based data analyst for non-technical users: upload data, get charts and stats	Fast exploratory analysis
Hex, Deepnote	Collaborative AI notebooks mixing SQL, Python and charts	Team analytics
Gamma, Canva, Copilot in PowerPoint	AI presentations from prompts or documents	Turning analysis into slides

Core data stack

Category	Tools	Learn first
Spreadsheets	Excel (365), Google Sheets	Excel

Category	Tools	Learn first
Databases & SQL	SQL Server + SSMS, MySQL, PostgreSQL, SQLite, DuckDB; cloud: Snowflake, BigQuery, Databricks, Microsoft Fabric, Amazon Redshift	SQL Server or PostgreSQL
Programming	Python (pandas, NumPy, matplotlib/seaborn/plotly, scikit-learn, statsmodels), R	Python
Notebooks / IDEs	Jupyter, Google Colab, VS Code, PyCharm	Jupyter + VS Code
BI	Power BI, Tableau, Looker Studio (free), Metabase (open source), Qlik	Power BI (and Tableau)
Data engineering	Power Query, dbt, Airflow, Azure Data Factory, Fabric pipelines	Power Query
ML / MLOps	scikit-learn, XGBoost, TensorFlow/PyTorch, MLflow, Azure ML, Vertex AI, SageMaker	scikit-learn
Version control	Git, GitHub	GitHub for your portfolio
Automation	Power Automate, Zapier, Make, n8n, Windows Task Scheduler, cron	Power Automate
Collaboration	Jira, Confluence, Notion, Teams, Slack	Jira basics

Safe & smart use

✔ Use your company's approved AI tools for work data; personal accounts may use your inputs for training unless you opt out

✔ Mask personal data (names, phone, PAN, Aadhaar) before uploading

✔ Recalculate any number you will present; ask the tool to show its code

✔ Ask for sources (Perplexity, deep research) and open them

✔ Keep prompts that work in a personal library

✔ Compare two assistants on important questions; they make different mistakes

✔ Declare AI assistance where your organisation or course requires it

Resources

- ▶ YouTube: Simplilearn ChatGPT tutorial for beginners · ▶ YouTube: Simplilearn AI tools for data analysis
- ▶ YouTube: Claude AI tutorial for data analysis · ▶ YouTube: Perplexity AI tutorial for research · ▶ YouTube: NotebookLM tutorial
- ▶ YouTube: Lovable AI tutorial build app · ▶ YouTube: Microsoft Copilot in Excel tutorial · ▶ YouTube: GitHub Copilot for data science
- 📘 ChatGPT · 📘 Claude · 📘 Gemini · 📘 Microsoft Copilot · 📘 Grok · 📘 Perplexity
- 📘 Lovable · 📘 NotebookLM · 📘 Cursor · 📘 GitHub Copilot · 📘 Julius AI

- 📘 CompareGen: AI tools for data analysis 2026